

HelixVPN — Requirements Traceability Matrix (requirement → component → test)

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Revision: 8 Last modified: 2026-07-06T10:44:05Z **Status:** active
— Volume 0 (Spine, meta & governance) nano-detail document

Rev 8 (GAP-4 closure, 2026-07-06): Created `v04-client/connector.md` as the single consolidating nano-detail owner for the Connector requirement band (FR-701..707) and updated the FR-7xx owning-doc column to point to it. Also repointed NFR-603 to `connector.md` as its primary owning doc. Updated §6 GAP-4 to **CLOSED**.

Rev 7 (GAP-3 doc-level closure + DDoS targets, 2026-07-06): Closed GAP-3 at the documentation level (`v06-deploy/disaster-recovery.md` consolidates source-coverage ledger gap G1 and pins RTO/RPO targets/runbooks; the RTO/RPO numbers remain measurement-pending until the first Phase-2 CHAOS region-failover drill). Pinned quantitative DDoS design targets for NFR-413/NFR-414 in `v08-testing/ddos.md` §10 and updated the coverage-ledger GAP-6 closure statement to reference them.

Rev 6 (GAP-5 doc-level closure): Updated §6 GAP-5 to **CLOSED at doc level** — all 15 `v08-testing/` nano-detail docs are authored and on disk, the `coverage-ledger-schema.md` is authored, and every FR/NFR maps to a planned §11.4.169 test type. The residual

"evidence state PENDING" is the honest spec-phase state (no code built yet), not a documentation gap. Added the v08-testing/source-doc list to §8.

Rev 5 (Phase-1 consolidation pass): Closed GAP-6 in source docs by syncing the matrix with nonfunctional-requirements.md Rev 2 (NFR-411..414 added there) and the new FR-610 added to functional-requirements.md Rev 4. Added traceability rows for NFR-411/412/413/414 and NFR-700/701/702/703, plus FR-609/610. Updated §6 GAP-6 to CLOSED. No existing row semantics changed. **Rev 4:** Independent gap-analysis pass (enterprise-hardening audit). Added **GAP-6** — the DDoS test-type tag is defined in the §1 legend but traced to **zero** FR/NFR rows, i.e. the API gateway/edge has no requirement-level rate-limiting/DDoS-resilience obligation despite 04_ARCH §4.6 naming a token-bucket mechanism; also flagged RBAC's lone-parenthetical status on FR-601. Added a §0 Mermaid diagram visualizing the Req → Component → Test type → Ledger row → Evidence state chain the prose already describes, to make the loop legible in one look. No existing row changed; this document still does not mint new requirement ids (that stays FR/NFR-doc territory). **Rev 3:** §1 legend — marked **UI/UX/REC** and **META** as *adjacent* types (NOT members of the §11.4.169 13-type set), added a member column; reworded the "Sources verified" MASTER_INDEX citation to cite on-disk existence ([DONE]) rather than a "to-be-generated" status as existence proof (§11.4.6). **Rev 2:** Corrected GAP-3/GAP-5 — v06-deploy/disaster-recovery.md and all 15 v08-testing/ docs are now authored (the stale "not-yet-authored" framing fixed per §11.4.6; residual gaps are pending-measurement, not missing docs). **Authority:** Subordinate to ../SPECIFICATION.md. Requirement ids are owned by ../v01-product/functional-requirements.md (HVPN-FR-NNN) and ../v01-product/nonfunctional-requirements.md (HVPN-NFR-NNN); this document does **not** mint new requirements — it cross-references them. Test types are the §11.4.169 closed set owned by ../10-testing-acceptance-and-qa.md.

Document role. The closed-loop cross-reference matrix tying every HelixVPN requirement (HVPN-FR-NNN / HVPN-NFR-NNN) to (1) the **satisfying component** — the Volume 2-6/10 spec doc that implements it — and (2) the **§11.4.169 test type(s) + coverage-ledger row** that prove it. It closes the FR → component → test → evidence loop the FR/NFR docs declare. Per §11.4.6, any requirement with **no** satisfying component or **no** test type is flagged a **GAP** in §6 (honest, never hidden). Per §11.4.118, §7 states the enumerated coverage this matrix claims — and the boundary of that claim.

SPEC-PHASE HONESTY (§11.4.6 / §11.4.108). HelixVPN is in specification, not build. Every "evidence-state" in this matrix is therefore **PENDING** (designed, not yet captured): the coverage-ledger rows are *planned* rows whose state will

be MISSING → DESIGNED → AUTONOMOUS_DESIGNED → AUTONOMOUS_VERIFIED once the Volume-8 harness runs (v08-testing/coverage-ledger-schema.md). This document maps the *intended* test type per requirement; it asserts **no** captured PASS. A PASS is earned only when the named test produces captured evidence (§11.4.5/.69/.107).

Table of contents

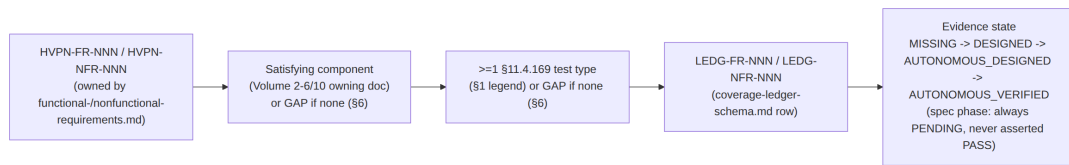
- [0. How to read a traceability row](#)
 - [1. The §11.4.169 test-type legend](#)
 - [2. Functional-requirement traceability \(HVPN-FR-NNN\)](#)
 - [3. Non-functional-requirement traceability \(HVPN-NFR-NNN\)](#)
 - [4. MVP Definition-of-Done → requirement → gate](#)
 - [5. Phase-0 gate → requirement coverage](#)
 - [6. GAP register \(§11.4.6 — honest\)](#)
 - [7. Coverage claim & boundary \(§11.4.118\)](#)
 - [Sources verified](#)
-

0. How to read a traceability row

Each FR/NFR is one row with five cross-reference fields:

Field	Meaning
Req ID	HVPN-FR-NNN / HVPN-NFR-NNN (owned by the FR/NFR docs; never minted here).
Satisfying component (owning doc)	the Volume 2–6/10 spec doc + component that implements it. GAP if none.
§11.4.169 test type(s)	the mandatory test type(s) whose captured evidence will prove it (closed set §1). GAP if none.
Coverage-ledger row	the planned feature × test-type × evidence-state ledger key (LEDG-FR-NNN / LEDG-NFR-NNN). State = PENDING in spec phase.
Priority / phase	MVP / P2 / P3 / P0 / ALWAYS (from the FR/NFR doc).

The ledger row id is derived 1:1 from the requirement id (LEDG-FR-001 ↔ HVPN-FR-001) so the Volume-8 coverage ledger (coverage-ledger-schema.md) can be populated mechanically from this matrix.



1. The §11.4.169 test-type legend

The closed mandatory test-type set is the **13** §11.4.169 types (the first 13 rows, 10-testing-acceptance-and-qa.md §2 / §5). **UI/UX/REC** and **META** are *adjacent* types (NOT members of the §11.4.169 13-type set) — UI/UX derives from §11.4.48/49 + §11.4.162 and recorded-evidence from §11.4.153/.158/.159; META is the §1.1 paired- mutation discipline applied to every gate. Each requirement names ≥ 1 §11.4.169 type (and, where user-visible, an adjacent UI/UX/REC pass):

Tag	Test type	§11.4.169 member	Volume-8 §5 home
U	unit	yes	§5.1
INT	integration (containers-submodule infra)	yes	§5.2
E2E	end-to-end (netns rig)	yes	§5.3
FA	full-automation (autonomous, re-runnable)	yes	§5.4
CHAL	Challenges (challenges-submodule banks)	yes	§5.5
HQA	HelixQA autonomous sessions	yes	§5.6
SEC	security	yes	§5.7
DDOS	DoS / load-flood	yes	§5.8
STR/ CHAOS	stress + chaos	yes	§5.9
CONC	concurrency / atomicity	yes	§5.10
RACE	race-condition / deadlock	yes	§5.11

Tag	Test type	§11.4.169 member	Volume-8 §5 home
MEM	memory (iOS NE soak)	yes	§5.12
BENCH/ PERF/ SCALE	benchmarking / performance / scale	yes	§5.13
UI/UX/ REC	UI/UX + recorded- evidence (vision_engine)	adjacent (§11.4.48/.49/.162/.153/.158/.159)	§5.14
META	paired §1.1 meta-test mutation	adjacent (§1.1)	every gate

Every row implicitly also carries **META** (the §1.1 paired mutation per gate) and, for any user-visible feature, a **CHAL/HQA** anti-bluff pass; the columns below name the *primary discriminating* type(s) to keep the matrix legible.

2. Functional-requirement traceability (HVPN-FR-NNN)

Satisfying-component owning docs are quoted from functional-requirements.md; the test types are assigned from each FR's acceptance criterion ([evidence]-flagged criteria require captured runtime evidence per §11.4.69).

A. Connect & transport (FR-0xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-001	WG crypto core — 01, v02-data-plane/wireguard-core.md	SEC (CI crypto-lint) + INT (handshake)	LEDG-FR-001	MVP
HVPN-FR-002	plain-UDP transport — v02-data-plane/transport-plain-udp.md	PERF/ BENCH (G1 iperf3) + E2E	LEDG-FR-002	MVP/ P0
HVPN-FR-003	Transport trait — 01 §7.1, v02-data-plane/transport-trait.md	U + INT (shared-crate parity)	LEDG-FR-003	MVP MVP

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-004	MASQUE/QUIC — v02-data-plane/transport-masque-quic.md	E2E (DPI sim) + PERF (G2)	LEDG-FR-004	MVP
HVPN-FR-005	LWO — v02-data-plane/transport-lwo.md	E2E (signature-block evasion)	LEDG-FR-005	
HVPN-FR-006	auto-ladder — v02-data-plane/transport-selection-ladder.md	E2E (DoD#4 escalation) + FA	LEDG-FR-006	
HVPN-FR-007	helix_pin_transport — 03, v04-client/helix-core-rust.md	U + INT	LEDG-FR-007	MVP
HVPN-FR-008	port evasion :443/:53/hop — transport-masque-quic.md + edge	E2E + INT	LEDG-FR-008	MVP
HVPN-FR-009	Shadowsocks-wrap — v02-data-plane/transport-shadowsocks.md	E2E	LEDG-FR-009	P2
HVPN-FR-010	UDP-over-TCP — v02-data-plane/transport-udp-over-tcp.md	E2E (total-UDP-block)	LEDG-FR-010	P2
HVPN-FR-011	per-network memory/regional priors — transport-selection-ladder.md	INT + FA	LEDG-FR-011	P2
HVPN-FR-012	CONNECT-IP — 01, transport-masque-quic.md	E2E (interop)	LEDG-FR-012	P2
HVPN-FR-013	full-tunnel privacy exit — 01, v04-client/helix-core-rust.md	E2E (egress-IP)	LEDG-FR-013	MVP
HVPN-FR-014	split tunneling — 03, helix-core-rust.md	E2E (per-route/per-app)	LEDG-FR-014	MVP
HVPN-FR-015	transient-drop recovery — v02-data-plane/orchestrator-and-state.md	CHAOS + E2E	LEDG-FR-015	MVP
				MVP

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-016	MTU mgmt — transport-plain-udp.md, wireguard-core.md	INT (path-MTU probe)	LEDG-FR-016	
HVPN-FR-017	DAITA — v02-data-plane/daita.md	BENCH (overhead) + E2E	LEDG-FR-017	P2
HVPN-FR-018	status stream — 03 §7.3, v04-client/state-management.md	U + UI/UX	LEDG-FR-018	MVP
HVPN-FR-019	per-leg topology — 01 (D6), svc-coordinator.md	INT	LEDG-FR-019	P2
HVPN-FR-020	P2P + NAT + relay — 01, routing-and-addressing.md	E2E (real NAT) + CHAOS	LEDG-FR-020	P2

B. Identity & enrollment (FR-1xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-101	OIDC — v03-control-plane/svc-identity.md	INT (OIDC login)	LEDG-FR-101	MVP
HVPN-FR-102	anon device token — svc-identity.md, identity-and-enrollment.md	INT (no-PII enroll) + SEC	LEDG-FR-102	MVP
HVPN-FR-103	on-device keygen, key-never-leaves — identity-and-enrollment.md, svc-pki.md	SEC (schema + wire capture)	LEDG-FR-103	MVP
HVPN-FR-104	cert+token-gated enroll — svc-identity.md, identity-and-enrollment.md	SEC + INT	LEDG-FR-104	MVP
HVPN-FR-105	short-lived mTLS device cert — svc-pki.md, pki-and-certs.md	SEC + INT (auto-renew)	LEDG-FR-105	MVP
		SEC		MVP

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-106	control/data key separation — 04 §1.2, svc-pki.md		LEDG-FR-106	
HVPN-FR-107	revoke < 1s — svc-registry.md, svc-pki.md, edge	PERF (revoke timing) + SEC	LEDG-FR-107	MVP
HVPN-FR-108	per-tenant CA — svc-pki.md, pki-and-certs.md	SEC (cross-tenant FAIL)	LEDG-FR-108	MVP
HVPN-FR-109	auto cert rotation no-drop — pki-and-certs.md	INT (rotate mid-session)	LEDG-FR-109	MVP
HVPN-FR-110	RLS tenant isolation — data-model-ddl.md, svc-identity.md	SEC (RLS bypass attempt)	LEDG-FR-110	MVP
HVPN-FR-111	all-roles enroll — svc-registry.md	INT (DoD#2)	LEDG-FR-111	MVP

C. Policy & authorization (FR-2xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-201	default-deny/fail-closed — svc-policy.md, zero-trust-and-default-deny.md	SEC + E2E + META	LEDG-FR-201	MVP
HVPN-FR-202	ACL compiler grammar — svc-policy.md	U (compile) + INT	LEDG-FR-202	MVP
HVPN-FR-203	AllowedIPs + verdict map — svc-policy.md, 01 §8	E2E (allow/deny flow)	LEDG-FR-203	MVP
HVPN-FR-204	authorized-reach + denied — svc-policy.md, edge	E2E (DoD#3)	LEDG-FR-204	MVP
HVPN-FR-205	policy edit < 1s no-restart — svc-policy.md, svc-coordinator.md	PERF (DoD#5) + E2E	LEDG-FR-205	MVP
		E2E + SEC		MVP

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-206	split-horizon default — svc-policy.md, routing-and-addressing.md		LEDG-FR-206	
HVPN-FR-207	need-to-know map filtering — svc-coordinator.md, 04 (S3)	SEC (snapshot omits) + META	LEDG-FR-207	MVP
HVPN-FR-208	policy-as-code/ GitOps — svc-policy.md	INT (repo reconcile)	LEDG-FR-208	P2

D. Routing, addressing & multi-network (FR-3xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-301	1→N overlay — routing-and-addressing.md, svc-policy.md	E2E (≥ 2 nets)	LEDG-FR-301	MVP
HVPN-FR-302	overlapping-CIDR resolve — routing-and-addressing.md, svc-ipam.md	E2E (same-CIDR distinct)	LEDG-FR-302	MVP
HVPN-FR-303	ULA /48 + 4via6 (+NAT fallback) — svc-ipam.md, routing-and-addressing.md	E2E + INT	LEDG-FR-303	MVP
HVPN-FR-304	stable overlay IP — svc-ipam.md	INT (persist across reconnect)	LEDG-FR-304	MVP
HVPN-FR-305	connector advertise + route — svc-registry.md, edge, routing-and-addressing.md	E2E (DoD#3 path)	LEDG-FR-305	MVP
HVPN-FR-306	two-way no-inbound — 01, edge, connector spec	E2E + SEC (no-inbound)	LEDG-FR-306	MVP
HVPN-FR-307	IPAM pool alloc — svc-ipam.md	STR/CHAOS (concurrent	LEDG-FR-307	MVP

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
		enroll) + CONC		

E. Multi-hop (FR-4xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-401	nested WG per-hop keys — multihop.md	E2E (2-hop, egress=exit)	LEDG-FR-401	P2
HVPN-FR-402	control-plane path selection — multihop.md, svc-coordinator.md	INT + E2E	LEDG-FR-402	P2
HVPN-FR-403	jurisdiction separation — multihop.md	E2E (different regions)	LEDG-FR-403	P2

F. Kill-switch & leak protection (FR-5xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-501	core-owned kill-switch state — kill-switch-and-dns-leak.md	U (state machine) + SEC	LEDG-FR-501	MVP
HVPN-FR-502	no plaintext on drop/escalate — kill-switch-and-dns-leak.md	SEC/E2E (leak pcap, DoD#7)	LEDG-FR-502	MVP
HVPN-FR-503	DNS forced through tunnel — kill-switch-and-dns-leak.md	SEC/E2E (DNS-leak test)	LEDG-FR-503	MVP
HVPN-FR-504	per-OS firewall from shared SM — kill-switch-and-dns-leak.md, shims	SEC + per-OS branch	LEDG-FR-504	MVP

G. Console & administration (FR-6xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-601	CRUD entities — svc-api.md, web-console.md	INT + E2E (RBAC)	LEDG-FR-601	MVP
HVPN-FR-602	API-client-only build (no core_ffi) — web-console.md, 03	INT (per-flavor build)	LEDG-FR-602	MVP
HVPN-FR-603	live WS/SSE no-poll — svc-api.md	E2E (live push)	LEDG-FR-603	MVP
HVPN-FR-604	live topology view — web-console.md	UI/UX + E2E	LEDG-FR-604	MVP
HVPN-FR-605	control-action audit only — audit-and-compliance.md, svc-telemetry.md	SEC (schema+UI, no-flow)	LEDG-FR-605	MVP
HVPN-FR-606	multi-tenant isolation — web-console.md, RLS	SEC	LEDG-FR-606	MVP
HVPN-FR-607	optional billing — web-console.md	INT	LEDG-FR-607	P3
HVPN-FR-608	responsive light+dark Console — web-console.md, Volume 10	UI/UX (visual-regression)	LEDG-FR-608	MVP
HVPN-FR-609	exportable control-action audit slice — audit-and-compliance.md, svc-telemetry.md	SEC + INT	LEDG-FR-609	MVP
HVPN-FR-610	RBAC role→action matrix enforcement — svc-identity.md, svc-api.md	SEC + INT + E2E	LEDG-FR-610	MVP

H. Connector (FR-7xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-701	<u>v04-client/connector.md</u> (secondary:	SEC (no-inbound) + E2E	LEDG-FR-701	MVP

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-702	helix-core-rust.md §9.1) v04-client/connector.md (secondary: helix-core-rust.md §9.2 , shim-linux.md §9)	INT	LEDG-FR-702	MVP
HVPN-FR-703	v04-client/connector.md (secondary: helix-core-rust.md §9.1)	INT (shared-crate)	LEDG-FR-703	MVP
HVPN-FR-704	v04-client/connector.md (secondary: svc-registry.md , edge)	E2E	LEDG-FR-704	MVP
HVPN-FR-705	v04-client/connector.md (secondary: svc-policy.md §4.2 , helix-core-rust.md §9.3)	INT (local-ACL honoured + central-deny wins)	LEDG-FR-705	MVP
HVPN-FR-706	v04-client/connector.md (secondary: helix-core-rust.md §9.2 , shim-android.md)	E2E (embedded target)	LEDG-FR-706	P2
HVPN-FR-707	v04-client/connector.md traces availability-following to v02-data-plane/orchestrator-and-state.md	CHAOS (drop→resume)	LEDG-FR-707	MVP

I. Observability & telemetry (FR-8xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-801	no durable connection table — no-logging-as-code.md, data-model-ddl.md	SEC (schema-lint, DoD#8) + META	LEDG-FR-801	MVP
HVPN-FR-802	ephemeral TTL presence — svc-events.md, no-logging-as-code.md	INT (TTL expiry)	LEDG-FR-802	MVP
HVPN-FR-803	Prometheus counters/health — svc-telemetry.md, observability.md	INT + SEC (no-flow labels)	LEDG-FR-803	MVP
HVPN-FR-804	convergence/event-lag SLO metrics — observability.md, NFR doc	PERF	LEDG-FR-804	MVP
HVPN-FR-805	counts-only success metrics — svc-telemetry.md, success-metrics.md	U (label audit) + SEC	LEDG-FR-805	MVP

J. Deployment & self-host (FR-9xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-901	helixvpctl init from zero — helixvpctl.md, podman-quadlets.md	E2E/FA (DoD#1)	LEDG-FR-901	MVP
HVPN-FR-902	rootless Podman (§11.4.161) — podman-quadlets.md	INT/SEC (no-root)	LEDG-FR-902	MVP
HVPN-FR-903	same images scale to HA — ha-and-multiregion.md	INT (multi-region)	LEDG-FR-903	P2
HVPN-FR-904	helixvpctl subcommands — helixvpctl.md	INT/E2E	LEDG-FR-904	MVP
HVPN-FR-905	Compose + K8s manifests — docker-	INT (stack bring-up)	LEDG-FR-905	P2

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
	compose.md, kubernetes.md			
HVPN-FR-906	hardened edge container — podman-quadlets.md, 04 (S8)	SEC	LEDG-FR-906	MVP
HVPN-FR-907	fail-static — edge, 01, 02	CHAOS (kill control plane)	LEDG-FR-907	MVP
HVPN-FR-908	§11.4.77 regen mechanism — repo-layout-and-decoupling.md	INT (fresh-clone bootstrap)	LEDG-FR-908	MVP

K. Clients & platform apps (FR-10xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-1001	runHelixApp 3 flavors — helix-ui-flutter.md	INT (per-flavor build)	LEDG-FR-1001	MVP
HVPN-FR-1002	shared core, UI=f(stream) — helix-core-rust.md, ffi-surface.md, state-management.md	INT + UI/UX	LEDG-FR-1002	MVP
HVPN-FR-1003	one-button connect — helix-ui-flutter.md, Volume 10	UI/UX + E2E	LEDG-FR-1003	MVP
HVPN-FR-1004	iOS NE memory ceiling +≥30% — shim-apple.md, 03	MEM (G3 on-device soak)	LEDG-FR-1004	MVP/P0
HVPN-FR-1005	Android VpnService+JNI — shim-android.md	E2E + CHAOS (background-kill)	LEDG-FR-1005	MVP
HVPN-FR-1006	Linux kernel WG+systemd — shim-linux.md	E2E	LEDG-FR-1006	MVP
HVPN-FR-1007	Windows wireguard-nt+service+WFP	E2E (per-OS)	LEDG-FR-1007	P2

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
	— shim-windows.md			
HVPN-FR-1008	macOS NE — shim-apple.md	E2E (per-OS)	LEDG-FR-1008	P2
HVPN-FR-1009	HarmonyOS NEXT — shim-harmonyos.md	E2E (on-device)	LEDG-FR-1009	P3
HVPN-FR-1010	Aurora OS — shim-aurora.md	E2E (on-device)	LEDG-FR-1010	P3
HVPN-FR-1011	Web = Console + optional WASM proxy — web-console.md	E2E + documentation-honesty	LEDG-FR-1011	P3
HVPN-FR-1012	pick exit/network + toggle obfs — helix-ui-flutter.md, ffi-surface.md	UI/UX + E2E	LEDG-FR-1012	MVP
HVPN-FR-1013	light+dark all clients — Volume 10	UI/UX (visual-regression)	LEDG-FR-1013	MVP
HVPN-FR-1014	all three apps drive system — all client docs	E2E/FA (DoD#8)	LEDG-FR-1014	MVP

L. Post-quantum & advanced privacy (FR-11xx)

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
HVPN-FR-1101	ML-KEM PSK handshake — post-quantum.md	SEC + INT (interop+measure)	LEDG-FR-1101	P2
HVPN-FR-1102	hybrid never PQ-only — post-quantum.md	SEC (PQ-off still secure)	LEDG-FR-1102	P2
HVPN-FR-1103	Rosenpass evaluation —	(documented evaluation) — see GAP-2	LEDG-FR-1103	P2

Req ID	Satisfying component (owning doc)	§11.4.169 test type(s)	Ledger row	Phase
	post-quantum.md			

3. Non-functional-requirement traceability (HVPN-NFR-NNN)

Each NFR already names its **Verify by** test type in nonfunctional-requirements.md; this table adds the satisfying-component owning doc + ledger row. UNVERIFIED targets are the NFR doc's; this matrix does not assert them met.

Req ID	Statement (short)	Satisfying component (owning doc)	§11.4.169 test type	Ledger row	Phase
HVPN-NFR-001	plain-UDP ≥80% bare-link	transport-plain-udp.md	PERF/BENCH (G1)	LEDG-NFR-001	P0
HVPN-NFR-002	MASQUE ≥50% through DPI	transport-masque-quic.md	PERF/BENCH (G2)	LEDG-NFR-002	P0
HVPN-NFR-003	convergence p99 < 1s	svc-coordinator.md	PERF	LEDG-NFR-003	MV
HVPN-NFR-004	revoke < 1s	svc-pki.md, svc-coordinator.md, edge	PERF + SEC	LEDG-NFR-004	MV
HVPN-NFR-005	bounded handshake latency	wireguard-core.md	INT	LEDG-NFR-005	MV
HVPN-NFR-006	bounded ladder escalation	transport-selection-ladder.md	INT	LEDG-NFR-006	MV
HVPN-NFR-007	bounded datapath added latency	transport-*	BENCH	LEDG-NFR-007	P2
HVPN-NFR-008	MTU no-fragment	transport-plain-udp.md, wireguard-core.md	INT	LEDG-NFR-008	MV
HVPN-NFR-009	admin policy-edit < 1s (superset)	svc-policy.md, svc-coordinator.md	E2E/FA	LEDG-NFR-009	MV

Req ID	Statement (short)	Satisfying component (owning doc)	§11.4.169 test type	Ledger row	Phase
HVPN-NFR-100	$\geq 10k$ streams/ coordinator	svc-coordinator.md	STR/ CHAOS (soak)	LEDG-NFR-100	
HVPN-NFR-101	coordinator memory slope ≈ 0	svc-coordinator.md	STR/ CHAOS + MEM	LEDG-NFR-101	MV
HVPN-NFR-102	minimal-affected-set fan-out	svc-coordinator.md	U + PERF	LEDG-NFR-102	MV
HVPN-NFR-103	many connectors/ tenant	svc-ipam.md	INT	LEDG-NFR-103	MV
HVPN-NFR-104	1 user $\rightarrow$ N nets	routing-and-addressing.md, svc-policy.md	E2E	LEDG-NFR-104	MV
HVPN-NFR-105	reconnect-storm absorbed	svc-coordinator.md, svc-events.md	STR (storm)	LEDG-NFR-105	MV
HVPN-NFR-106	per-tenant churn isolation	svc-coordinator.md	CONC + RACE	LEDG-NFR-106	MV
HVPN-NFR-107	stateless multi- replica coordinators	architecture-and-wiring.md, ha-and-multiregion.md	INT (multi- replica)	LEDG-NFR-107	P2
HVPN-NFR-108	bus swap Redis $\rightarrow$ NATS	svc-events.md, ha-and-multiregion.md	INT (bus swap)	LEDG-NFR-108	P2
HVPN-NFR-200	fail-static, 0 drops	edge, 01	CHAOS	LEDG-NFR-200	ALV
HVPN-NFR-201	DB/Redis blip no- restart	architecture-and-wiring.md	CHAOS	LEDG-NFR-201	MV
HVPN-NFR-202	Redis loss degrades gracefully	svc-events.md, no-logging-as-code.md	CHAOS	LEDG-NFR-202	MV
HVPN-NFR-203	no-work-loss XAutoClaim	svc-events.md	CHAOS	LEDG-NFR-203	MV
HVPN-NFR-204	coordinator restart transparent	svc-coordinator.md	CHAOS/INT	LEDG-NFR-204	MV
HVPN-NFR-205	region-failover RTO/ RPO (UNVERIFIED)	v06-deploy/ disaster-recovery.md	CHAOS (drill) — GAP-3 doc- level CLOSED; targets verified by	LEDG-NFR-205	P2

Req ID	Statement (short)	Satisfying component (owning doc)	§11.4.169 test type	Ledger row	Phase
			Phase-2 drill		
HVPN-NFR-206	gateway failover re-point	svc-coordinator.md, ha-and-multiregion.md	INT/CHAOS	LEDG-NFR-206	P2
HVPN-NFR-207	poison→DLQ	svc-events.md	CHAOS/INT	LEDG-NFR-207	MV
HVPN-NFR-300	no durable flow table	no-logging-as-code.md, data-model-ddl.md	SEC	LEDG-NFR-300	ALV
HVPN-NFR-301	schema-lint not a tautology	no-logging-as-code.md	META (§1.1)	LEDG-NFR-301	ALV
HVPN-NFR-302	bus carries no traffic shape	svc-events.md	SEC (payload-lint)	LEDG-NFR-302	MV
HVPN-NFR-303	audit = control-only	audit-and-compliance.md, svc-telemetry.md	U + INT	LEDG-NFR-303	MV
HVPN-NFR-304	ephemeral presence	svc-events.md	INT (TTL)	LEDG-NFR-304	MV
HVPN-NFR-305	coarse last_seen_at	no-logging-as-code.md	U + INT	LEDG-NFR-305	MV
HVPN-NFR-306	counts-only metric labels	svc-telemetry.md	U (label-cardinality)	LEDG-NFR-306	MV
HVPN-NFR-307	anonymous identity	svc-identity.md, identity-and-enrollment.md	INT	LEDG-NFR-307	MV
HVPN-NFR-308	no-logging runtime signature	no-logging-as-code.md	E2E CHAL (deployed-DB)	LEDG-NFR-308	MV
HVPN-NFR-400	WG crypto never forked	wireguard-core.md	SEC + META	LEDG-NFR-400	ALV
HVPN-NFR-401	outbound-only edges	01, edge	SEC	LEDG-NFR-401	ALV
HVPN-NFR-402	need-to-know peer filtering	svc-coordinator.md	E2E + META	LEDG-NFR-402	MV
HVPN-NFR-403	mTLS + instant revoke	pki-and-certs.md	SEC	LEDG-NFR-403	MV
HVPN-NFR-404	kill-switch 0 leak	kill-switch-and-dns-leak.md	SEC/E2E	LEDG-NFR-404	MV
			SEC/E2E		MV

Req ID	Statement (short)	Satisfying component (owning doc)	§11.4.169 test type	Ledger row	Phase
HVPN-NFR-405	DNS-leak 0 plaintext	kill-switch-and-dns-leak.md		LEDG-NFR-405	
HVPN-NFR-406	WG private key never leaves	identity-and-enrollment.md	INT + SEC	LEDG-NFR-406	MV
HVPN-NFR-407	hybrid PQ	post-quantum.md	SEC + INT	LEDG-NFR-407	P2
HVPN-NFR-408	RLS at DB	data-model-ddl.md	SEC/INT	LEDG-NFR-408	MV
HVPN-NFR-409	anti-bluff gauntlet per change	Volume 8	META (per-gate mutation)	LEDG-NFR-409	ALV
HVPN-NFR-410	censorship-evasion ladder	transport-selection-ladder.md	E2E (DPI-sim)	LEDG-NFR-410	MV
HVPN-NFR-411	runtime secrets rotatable without rebuild	repo-layout-and-decoupling.md, data-model-ddl.md	SEC + INT	LEDG-NFR-411	MV
HVPN-NFR-412	WG/device + edge transport key rotation cadence	pki-and-certs.md, wireguard-core.md	SEC + INT	LEDG-NFR-412	MV
HVPN-NFR-413	control-plane API rate limiting	svc-api.md, architecture-and-wiring.md	STRESS + SEC	LEDG-NFR-413	MV
HVPN-NFR-414	data-plane edge DDoS/UDP-flood/amplification resilience	svc-api.md (enroll limits), edge (v02-data-plane/routing-and-addressing.md / transport-masque-quic.md)	DDOS	LEDG-NFR-414	MV
HVPN-NFR-500	iOS NE +≥30% headroom	shim-apple.md	MEM (G3)	LEDG-NFR-500	P0
HVPN-NFR-501	battery-sensitive push	orchestrator-and-state.md, shim-*	BENCH (battery)	LEDG-NFR-501	MV
HVPN-NFR-502	small core footprint	helix-core-rust.md	BENCH (artifact-size)	LEDG-NFR-502	MV
HVPN-NFR-503	≤60% host RAM (§12.6)	Volume 8 harness	BENCH (host sampler)	LEDG-NFR-503	ALV

Req ID	Statement (short)	Satisfying component (owning doc)	§11.4.169 test type	Ledger row	Phase
HVPN-NFR-504	control-plane bounded mem	svc-coordinator.md	STR/CHAOS + MEM	LEDG-NFR-504	MV
HVPN-NFR-505	lightweight samplers	observability.md	BENCH	LEDG-NFR-505	MV
HVPN-NFR-506	slow-consumer drop not buffer	svc-coordinator.md	STR + MEM	LEDG-NFR-506	MV
HVPN-NFR-600	one Flutter tree 3 flavors	helix-ui-flutter.md	INT	LEDG-NFR-600	MV
HVPN-NFR-601	core byte-for-byte shared	transport-trait.md, helix-core-rust.md	INT (shared-core parity)	LEDG-NFR-601	MV
HVPN-NFR-602	MVP iOS/Android/Linux	shim-apple.md, shim-android.md, shim-linux.md	E2E/FA	LEDG-NFR-602	MV
HVPN-NFR-603	connector headless Linux/Win/macOS	<u>v04-client/connector.md</u> (secondary: shim-linux.md §9, shim-windows.md, shim-apple.md)	INT	LEDG-NFR-603	MV
HVPN-NFR-604	Console web+desktop	web-console.md	E2E (responsive)	LEDG-NFR-604	MV
HVPN-NFR-605	P2 Windows+macOS	shim-windows.md, shim-apple.md	E2E	LEDG-NFR-605	P2
HVPN-NFR-606	P3 HarmonyOS+Aurora (UNVERIFIED)	shim-harmonyos.md, shim-aurora.md	E2E (per-shim)	LEDG-NFR-606	P3
HVPN-NFR-607	Web honestly scoped	web-console.md	E2E + doc-honesty	LEDG-NFR-607	P3
HVPN-NFR-608	light+dark visual-regression	Volume 10	UI/UX	LEDG-NFR-608	P3
HVPN-NFR-609	cross-platform parity (§11.4.81)	per-OS test branches	per-OS branch	LEDG-NFR-609	ALV
HVPN-NFR-700	zero-downtime rolling upgrades	ha-and-multiregion.md, architecture-and-wiring.md	CHAOS + INT	LEDG-NFR-700	P2
HVPN-NFR-701	backward-compatible Postgres migrations	data-model-ddl.md, codegen-pipeline.md	INT	LEDG-NFR-701	P2
HVPN-NFR-702			U + META	LEDG-NFR-702	MV

Req ID	Statement (short)	Satisfying component (owning doc)	§11.4.169 test type	Ledger row	Pha
	protobuf/OpenAPI wire-compatibility window	protobuf-spec.md, codegen-pipeline.md			
HVPN-NFR-703	infra sizing/cost forecast formula	observability.md, ha-and-multiregion.md	BENCH	LEDG-NFR-703	P2

4. MVP Definition-of-Done → requirement → gate

The 8-criteria MVP DoD (functional-requirements.md §M, [SPEC §8.1]) is the MVP acceptance checklist; each criterion maps to FRs whose ledger rows must reach AUTONOMOUS_VERIFIED with captured evidence.

DoD #	Criterion	FRs	Primary test type	NFR SLO
DoD-1	self-host from zero	FR-901, FR-902	E2E/FA	—
DoD-2	enroll connector + client	FR-104, FR-111, FR-701	INT/SEC	—
DoD-3	authorized-reach + denied	FR-204, FR-201, FR-305, FR-306	E2E	—
DoD-4	auto-escalate to MASQUE	FR-006, FR-004	E2E	NFR-410
DoD-5	policy edit < 1s no-restart	FR-205	PERF	NFR-003/009
DoD-6	revoke < 1s	FR-107	PERF/SEC	NFR-004
DoD-7	kill-switch + DNS-leak	FR-501..504	SEC/E2E	NFR-404/405
DoD-8	no durable log + all 3 apps	FR-801, FR-1014	SEC + E2E/FA	NFR-300/308

5. Phase-0 gate → requirement coverage

The Phase-0 spike gates (G1–G6, [SPEC §8.0]) are the earliest evidence; they prove the make-or-break requirements *and resolve the spine decisions* (see [decision-register.md §5](#)).

Gate	Requirement(s) proven	Decision resolved
G1	FR-002, NFR-001	(datapath floor)
G2	FR-004, NFR-002	D1
G3	FR-1004, NFR-500	D2 (make-or-break)
G4	(edge benchmark)	D5
G5	FR-018 (FFI), FR-1002	D-FFI-*
G6	FR-015/reconcile (push)	(validates P4 push-not-poll)

6. GAP register (§11.4.6 — honest)

Surfaced, never hidden. A GAP = a requirement whose satisfying component or test is *not yet fully pinned*, or a coverage seam this matrix exposes. GAP-1 through GAP-5 are not missing *requirements* (the FR/NFR set is enumerated for those); each is a *pin-the-detail* item tracked as a §11.4.93 workable item. **GAP-6 is the one exception** — it is the inverse seam: a test type (DDOS) with zero requirements mapped to it, which this independent audit pass surfaces as a genuinely missing requirement recommendation (not silently assumed covered).

- **GAP-1 — FR-705 (connector local-ACL × central policy interaction) — CLOSED in source docs (2026-07-05).** The precedence rule (local-deny overrides central-allow; central-deny overrides local-allow; compiled output = central policy minus local-deny; connector advertises `local_denylist` to the coordinator) is now pinned and backported into `v03-control-plane/svc-policy.md §2/§4/§5/§7/§8/§10` and `v04-client/helix-core-rust.md §9.3`. **Owner:** `v03-control-plane/svc-policy.md` + `v04-client/helix-core-rust.md`. **Test:** INT (local-ACL honoured + central-deny wins).
- **GAP-2 — FR-1103 (Rosenpass) is an *evaluation*, not a built capability.** Its acceptance criterion is "a documented evaluation exists", so it has **no runtime test type** (correctly — it is a decision-input doc, not a feature). Recorded so the absence of a runtime test is *intentional*, not an oversight. **Owner:** `v05-security/post-quantum.md`.
- **GAP-3 — NFR-205 / DR runbook RTO/RPO — CLOSED at doc level (2026-07-06).** `v06-deploy/disaster-recovery.md` consolidates the [03_ZAI] RTO/RPO budget, KMS-encrypted backups, and Terraform-driven region-failover/restore

runbooks, closing source-coverage ledger gap G1 at the **documentation** level. The residual item is an explicit **measurement pending** note: the §1 RTO/RPO numbers are stated as **TARGETS** (§11.4.6), not measured baselines, and will be verified by the first **CHAOS region-failover drill in Phase 2** (NFR-205's release-blocking gate). Until that drill produces captured evidence, the targets stay UNVERIFIED and the coverage ledger honestly records the evidence state as PENDING. **Owner:** v06-deploy/disaster-recovery.md (G1 closed; numbers pending the Phase-2 drill).

- **GAP-4 — Connector has no single owning nano-detail doc. — CLOSED (2026-07-06).** `v04-client/connector.md` now consolidates FR-701..707 and is the primary owning doc for the Connector requirement band; secondary implementation authority remains in `helix-core-rust.md` §9, `svc-registry.md`, `routing-and-addressing.md`, and the Volume-4 shims. The traceability rows for FR-701..707 have been updated to point to `connector.md`.
- **GAP-5 — v08-testing/ docs and coverage-ledger schema authored — CLOSED at doc level.** Volume 8's 15 nano-detail docs (the 13 §11.4.169 per-type docs + `coverage-ledger-schema.md` + `test-rig.md`) now EXIST on disk; the LEDG-* ledger rows trace to `coverage-ledger-schema.md` (authored) and the per-type harness docs. Every FR/NFR maps to ≥ 1 planned §11.4.169 test type. The residual "evidence state PENDING" is the honest spec-phase state: no code is built, so no test has produced a captured PASS — that is a *build-phase* pending state, not a documentation gap. **Owner:** `docs/research/mvp/final/v08-testing/*` (authored); future implementation phases (P0-P3) will move evidence states from PENDING to AUTONOMOUS_VERIFIED.
- **GAP-6 — DDOS/RBAC/rate-limiting traceability gap — CLOSED in source docs (2026-07-05).** The original finding (Rev 4) was that the DDOS §11.4.169 type had zero requirement owners and RBAC was only parenthetical. The fix has been applied:
 - `v01-product/nonfunctional-requirements.md` already added **NFR-413** (control-plane API rate limiting) and **NFR-414** (data-plane edge DDoS/UDP-flood/amplification resilience, Verify by: DDOS).
 - `v01-product/functional-requirements.md` added **FR-610** (RBAC role→action matrix enforcement).
 - This matrix now lists LEDG-NFR-413, LEDG-NFR-414, and LEDG-FR-610 with their owning docs and test types.
 - **Residual:** NFR-413/414 quantitative targets remain UNVERIFIED until Phase-2 benchmarks run; this is honest per §11.4.6, not a missing requirement.

No requirement is orphaned of a component. Every FR/NFR above names ≥ 1 satisfying owning doc (GAP-4 is now closed with `v04-client/connector.md` as the consolidated Connector owner). Every FR/NFR names ≥ 1 §11.4.169 test

type *except* FR-1103 (intentionally an evaluation, GAP-2). The GAPs are pin-the-detail / doc-locality / pending-measurement seams (all owning docs now authored), all tracked, none a hidden hole (§11.4.118).

7. Coverage claim & boundary (§11.4.118)

What this matrix claims. Every enumerated requirement in `functional-requirements.md` (HVPN-FR-001...1014, 14 capability bands A-L, now including FR-609/610) and `nonfunctional-requirements.md` (HVPN-NFR-001...414 and HVPN-NFR-700...703, 8 families) is mapped to a satisfying component **and** a §11.4.169 test type **and** a LEDG-* ledger row, with the §6 GAP register surfaced (GAP-1 and GAP-6 closed in source docs as of 2026-07-05; GAP-3, GAP-4 and GAP-5 closed at doc level as of 2026-07-06).

What it does NOT claim (§11.4.6 honest boundary).

1. **No captured PASS.** Spec phase → every evidence-state is **PENDING**; this is an intent map, not a result. A requirement is "met" only when its named test produces captured evidence (§11.4.5/.69/.107) and its ledger row reaches `AUTONOMOUS_VERIFIED`.
2. **Not provably exhaustive.** A capability a later Volume-2-6 doc introduces that the overview did not enumerate becomes a *new* FR via the §11.4.93 workable-item path — this matrix covers the *enumerated* set, which is necessary, not provably complete (the §11.4.118 unknown-unknown boundary).
3. **UNVERIFIED targets stay unproven.** Every quantitative target the FR/NFR docs mark UNVERIFIED ($\geq 80\%$ / $\geq 50\%$ throughput, p99 < 1s, iOS NE headroom, RTO/RPO, ML-KEM sizes) is reproduced here as the *named gate's* to prove, never asserted met.

The reverse loop closes mechanically: LEDG-FR-NNN ↔ HVPN-FR-NNN ↔ owning doc ↔ §11.4.169 type, so the Volume-8 coverage ledger (`coverage-ledger-schema.md`) and the §11.4.93 workable-items DB can both be populated from this single matrix.

Sources verified

- `../v01-product/functional-requirements.md` — the FR ids, statements, acceptance criteria, owning-doc column, priority, §M DoD map, §N parity-matrix map, and the FR-705 pinned

local-ACL × central-policy precedence contract (GAP-1 CLOSED).

- [../v01-product/nonfunctional-requirements.md](#) — the NFR ids, targets, the **Verify by** §11.4.169 test type column, priority, §9 NFR→principle map, §10 convergence chain, and the UNVERIFIED/TARGET markers (G1/G2/G3/iOS-NE/RTO).
- [../10-testing-acceptance-and-qa.md](#) §2/§5 — the closed §11.4.169 test-type taxonomy (the §1 legend), §6 the coverage-ledger model (PENDING evidence-state), §7 per-phase acceptance gates.
- [../SPECIFICATION.md](#) §7 (cross-cutting contracts → FR-003/018/205), §8 (phase gates G1-G6 + 8-criteria DoD → §4/§5), §9 (decisions → decision-register.md).
- [../MASTER_INDEX.md](#) Volumes 2-10 (owning-doc filenames) + the on-disk existence (now [DONE]) of the v08-testing/ docs and v06-deploy/disaster-recovery.md — verified as files present on disk, not inferred from a "to-be-generated" status (GAP-3, GAP-5).
- [../99-source-coverage-ledger.md](#) gap G1 (DR/RTO consolidation → GAP-3).
- 04_VPN_CLD/HelixVPN-Architecture-Refined.md §4.6 (Redis usage — token-bucket rate limiting named but untraced → GAP-6, Rev 4).
- [glossary.md](#) (Rev 3 **API Gateway / Rate limiting / RBAC / Tenant** entries, cross-referencing this document's GAP-6).

Constitution bindings: §11.4.44 (revision header), §11.4.6 (no captured PASS asserted; UNVERIFIED targets owned by named gates; GAPs surfaced not hidden), §11.4.108 (evidence-state is the runtime-signature dimension — PENDING until captured on a clean deployment), §11.4.118 (enumerated-coverage claim + its boundary in §7), §11.4.169 (every requirement → a mandatory test type), §11.4.93 (each requirement + GAP → a workable item), §11.4.25 (coverage-ledger discipline), §11.4.65/.153 (HTML+PDF[+DOCX] exports follow in refinement).